

Band-Limitedness of the Warped Pullback of the Hardy Z -Function

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Abstract

Let Z denote the Hardy Z -function and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The warped pullback

$$Y(u) := Z(\Theta^{-1}(u)) / \sqrt{\Theta'(\Theta^{-1}(u))}$$

has windowed Fourier transform $\widehat{Y}_T$ satisfying $|K_T(\mu)|^2 / \Theta(T) = O_\mu(T^{-1/2})$ for every fixed $|\mu| > 1$, where $K_T = \widehat{Y}_T / (2\pi)$ and $\Theta(T) = \frac{1}{2}T \log(T/2\pi) + O(T)$. The proof uses Gabcke's second-order Riemann–Siegel remainder bound $R_{\text{RS}}(t) = O(t^{-3/4})$. Consequently the windowed second-moment spectral measures $d\rho_T(\mu) = |\widehat{Y}_T(\mu)|^2 / (2\pi\Theta(T)) d\mu$ concentrate on $[-1, 1]$ with total mass tending to 2, and any weak-* subsequential limit ρ has $\text{supp } \rho \subseteq [-1, 1]$.

1 Setup

Definition 1 (Hardy function). $\vartheta(t) := \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ (continuous branch, $\vartheta(0) = 0$), and $Z(t) := e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$, real-valued on $\mathbb{R}$.

Definition 2 (Warp). Fix $c > c_\star := -\inf_{t \in \mathbb{R}} \vartheta'(t)$. Set $\Theta(t) := \vartheta(t) + ct$. Then $\Theta'(t) = \vartheta'(t) + c \geq c - c_\star > 0$ on $\mathbb{R}$, so $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ diffeomorphism.

Definition 3 (Warped pullback). $Y(u) := Z(\Theta^{-1}(u)) / \sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal branch of the positive square root.

Definition 4 (Windowed Fourier transform). For $T \geq 2$ and $\mu \in \mathbb{R}$,

$$\widehat{Y}_T(\mu) := \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du, \quad K_T(\mu) := \widehat{Y}_T(\mu) / (2\pi).$$

Axioms used. **(RS)** Riemann–Siegel with Gabcke remainder [2]: $Z(t) = 2 \sum_{n \leq N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R_{\text{RS}}(t)$ with $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ and

$$R_{\text{RS}}(t) = O(t^{-3/4}),$$

obtained by retaining the second-order term of the Riemann–Siegel asymptotic series. Gabcke's 1979 thesis establishes the bound with an explicit effective constant.

(HL) Hardy–Littlewood: $\int_0^T |Z|^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T)$.

(ϑ) $\vartheta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$; $\vartheta''(t) = 1/(2t) + O(t^{-3})$.

2 Plancherel mass

Lemma 5 (Plancherel via warp). $\int_0^{\Theta(T)} |Y(u)|^2 du = \int_0^T |Z(t)|^2 dt$, hence

$$\int_{\mathbb{R}} |\widehat{Y}_T(\mu)|^2 d\mu = 2\pi \int_0^T |Z|^2 dt.$$

Proof. Change of variables $u = \Theta(t)$, $du = \Theta'(t) dt$: $\int |Y|^2 du = \int (|Z|^2/\Theta') \Theta' dt = \int |Z|^2 dt$. The second identity is Plancherel on $L^2([0, \Theta(T)])$. $\square$

Corollary 6 (Total mass). $\frac{1}{2\pi\Theta(T)} \int_{\mathbb{R}} |\widehat{Y}_T(\mu)|^2 d\mu \rightarrow 2$ as $T \rightarrow \infty$, where $\Theta(T) = \frac{1}{2}T \log(T/2\pi) + O(T)$.

Proof. Combine Lemma 5 with (HL): $\int_0^T |Z|^2 dt = T \log(T/2\pi) + O(T) \sim 2\Theta(T)$. $\square$

3 Mode factorization

Lemma 7 (Riemann–Siegel factorization of $\widehat{Y}_T$). For $T \geq 2$, $\mu \in \mathbb{R}$,

$$\widehat{Y}_T(\mu) = \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \sum_{n=1}^{N(T)} \sum_{\sigma \in \{\pm 1\}} \mathcal{J}_{n,\sigma}(T, \mu) + \mathcal{R}_T(\mu),$$

$$\mathcal{J}_{n,\sigma}(T, \mu) := n^{-1/2} \int_{t_n}^T \sqrt{\Theta'(t)} e^{i\Phi_{n,\sigma,\mu}(t)} dt, \quad \Phi_{n,\sigma,\mu}(t) := \sigma(\vartheta(t) - t \log n) - \mu\Theta(t),$$

with $t_n := \max(2, 2\pi n^2)$.

Proof. Substitute $u = \Theta(t)$ in the Fourier integral: $du = \Theta'(t)dt$, $Y(u)du = Z(t)\Theta'(t)/\sqrt{\Theta'(t)} dt = Z(t)\sqrt{\Theta'(t)} dt$. Insert (RS) and expand $\cos = (e^i + e^{-i})/2$; the $\sigma = \pm 1$ index tracks the two exponentials. Modes $n > N(t)$ do not contribute at time t , giving the cutoff $t \geq 2\pi n^2$. $\square$

Lemma 8 (Phase derivative). Let $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t)$. Then for $t \geq t_n$:

$$(i) \quad \Phi'_{n,\sigma,\mu}(t) = \Theta'(t) (\sigma\omega_n(t) - \mu).$$

$$(ii) \quad \omega_n(t) \in [0, 1).$$

$$(iii) \quad \text{For } |\mu| \geq 1 + \epsilon, |\Phi'_{n,\sigma,\mu}(t)| \geq \epsilon \Theta'(t).$$

Proof. (i) Direct: $\Phi' = \sigma\vartheta' - \sigma \log n - \mu\Theta' = \Theta'(\sigma\omega_n - \mu)$. (ii) For $t \geq 2\pi n^2$, $\vartheta'(t) \geq \log n$, so $\omega_n \geq 0$; and $\Theta' - (\vartheta' - \log n) = c + \log n > 0$, so $\omega_n < 1$. (iii) $\sigma\omega_n \in [-1, 1)$, so $|\sigma\omega_n - \mu| \geq |\mu| - 1 \geq \epsilon$. $\square$

4 Band-limit bound

Lemma 9 (Single-mode off-band bound). Fix $\epsilon > 0$ and $|\mu| \geq 1 + \epsilon$. For each $1 \leq n \leq N(T)$ and $\sigma \in \{\pm 1\}$,

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq \frac{C_1}{\epsilon n^{1/2}} \left(\frac{1}{\sqrt{\Theta'(T)}} + \frac{1}{\sqrt{\Theta'(t_n)}} \right) + \frac{C_2}{\epsilon^2 n^{1/2}}.$$

Proof. Integration by parts with $e^{i\Phi} = (i\Phi')^{-1}(d/dt)e^{i\Phi}$:

$$\mathcal{J}_{n,\sigma} = n^{-1/2} \left[\frac{\sqrt{\Theta'}}{i\Phi'} e^{i\Phi} \right]_{t_n}^T - n^{-1/2} \int_{t_n}^T \frac{d}{dt} \left(\frac{\sqrt{\Theta'}}{i\Phi'} \right) e^{i\Phi} dt.$$

Boundary: $|\sqrt{\Theta'}/\Phi'| \leq 1/(\epsilon\sqrt{\Theta'})$ by Lemma 8(iii).

Integrated part: $\frac{d}{dt}(\sqrt{\Theta'}/\Phi') = \Theta''/(2\sqrt{\Theta'}\Phi') - \sqrt{\Theta'}\Phi''/\Phi'^2$, with $\Phi'' = \sigma\vartheta'' - \mu\vartheta''$ (since $\Theta'' = \vartheta''$) so $|\Phi''| \leq (1 + |\mu|)|\vartheta''|$. Both pieces are bounded by $C|\vartheta''|/(\epsilon^2\Theta'^{3/2})$; integrating, $\int_2^\infty |\vartheta''|/\Theta'^{3/2} dt \leq C \int_2^\infty dt/(t(\log t)^{3/2}) < \infty$. $\square$

Lemma 10 (Remainder bound via Gabcke). *Let $\mathcal{R}_T(\mu) := \int_0^T R_{RS}(t)\sqrt{\Theta'(t)}e^{-i\mu\Theta(t)} dt$. Uniformly in $\mu \in \mathbb{R}$,*

$$|\mathcal{R}_T(\mu)| \leq \int_0^T |R_{RS}(t)|\sqrt{\Theta'(t)} dt = O(T^{1/4}\sqrt{\log T}).$$

Proof. By (RS), $|R_{RS}(t)| = O(t^{-3/4})$ for $t \geq 2$, and $\sqrt{\Theta'(t)} = O(\sqrt{\log t})$. Triangle inequality on the integral defining $\mathcal{R}_T$ gives

$$|\mathcal{R}_T(\mu)| \leq C + C' \int_2^T t^{-3/4}\sqrt{\log t} dt = O(T^{1/4}\sqrt{\log T}),$$

uniformly in μ . $\square$

Theorem 11 (Band-limit). *For every $\mu \in \mathbb{R}$ with $|\mu| > 1$,*

$$\frac{|K_T(\mu)|^2}{\Theta(T)} = O_\mu\left(\frac{1}{T^{1/2}}\right) \rightarrow 0, \quad T \rightarrow \infty,$$

locally uniformly on $\{|\mu| > 1\}$.

Proof. Fix $\epsilon > 0$, $|\mu| \geq 1 + \epsilon$. By Lemma 9,

$$\sum_{n=1}^{N(T)} \sum_{\sigma} |\mathcal{J}_{n,\sigma}| \leq \frac{2C_1}{\epsilon} \sum_{n=1}^{N(T)} \frac{1}{n^{1/2}\sqrt{\Theta'(t_n)}} + \frac{2C_1}{\epsilon\sqrt{\Theta'(T)}} \sum_{n=1}^{N(T)} \frac{1}{n^{1/2}} + \frac{2C_2}{\epsilon^2} \sum_{n=1}^{N(T)} \frac{1}{n^{1/2}}.$$

Using $\Theta'(t_n) \asymp \log n$ for $n \geq 2$, $\Theta'(T) \asymp \log T$, $\sum_{n \leq N} n^{-1/2} = O(\sqrt{N})$, $\sum_{n \leq N} 1/(n^{1/2}\sqrt{\log n}) = O(\sqrt{N/\log N})$, the mode sum is $O_\epsilon(T^{1/4})$. Combined with the remainder bound $|\mathcal{R}_T(\mu)| = O(T^{1/4}\sqrt{\log T})$ of Lemma 10,

$$|\widehat{Y}_T(\mu)| = O_\epsilon(T^{1/4}\sqrt{\log T}).$$

Hence

$$\frac{|K_T(\mu)|^2}{\Theta(T)} = \frac{|\widehat{Y}_T(\mu)|^2}{(2\pi)^2\Theta(T)} = \frac{O_\epsilon(T^{1/2}\log T)}{T\log T/2} = O_\epsilon\left(\frac{1}{T^{1/2}}\right).$$

Uniformity on compact subsets of $\{|\mu| > 1\}$ is immediate from the uniform lower bound on $\epsilon = |\mu| - 1$. $\square$

5 Weak-* concentration of the spectral measure

Definition 12. $d\rho_T(\mu) := |\widehat{Y}_T(\mu)|^2 / (2\pi\Theta(T)) d\mu$.

Corollary 13. *The family $\{\rho_T\}_{T \geq 2}$ of nonnegative Borel measures on $\mathbb{R}$ has total mass $\rho_T(\mathbb{R}) \rightarrow 2$, and for every $\epsilon > 0$,*

$$\rho_T(\mathbb{R} \setminus [-1 - \epsilon, 1 + \epsilon]) \longrightarrow 0, \quad T \rightarrow \infty.$$

Consequently, any weak- subsequential limit ρ is a nonnegative even Borel measure with $\text{supp } \rho \subseteq [-1, 1]$ and $\rho([-1, 1]) = 2$.*

Proof. Total mass $\rho_T(\mathbb{R}) \rightarrow 2$ by Corollary 6. Evenness of ρ_T from $|\widehat{Y}_T(-\mu)| = |\widehat{Y}_T(\mu)|$ (reality of Y).

Middle off-band window. Fix $\epsilon > 0$ and $M > 1 + \epsilon$. By Theorem 11 with constants uniform on $[1 + \epsilon, M]$,

$$\rho_T(\{1 + \epsilon \leq |\mu| \leq M\}) \leq \int_{1+\epsilon \leq |\mu| \leq M} O_\epsilon(T^{-1/2}) d\mu = O_\epsilon\left(\frac{M}{T^{1/2}}\right).$$

Tail $|\mu| > M$ via iterated integration by parts in u . Since Z is entire and Θ is a C^∞ diffeomorphism with $\Theta' \geq c - c_\star > 0$, the function Y is C^∞ on $\mathbb{R}$. Integration by parts k times in u on the integral defining $\widehat{Y}_T(\mu) = \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$ gives

$$|\widehat{Y}_T(\mu)| \leq \frac{C_k(T)}{|\mu|^k}, \quad C_k(T) := \sum_{j=0}^{k-1} (|Y^{(j)}(0)| + |Y^{(j)}(\Theta(T))|) + \int_0^{\Theta(T)} |Y^{(k)}(u)| du.$$

Each $Y^{(j)}$ is bounded on $[0, \Theta(T)]$ by a polynomial in T : the convexity bound $|Z(t)| = O(t^{1/4+\delta})$ and its derivative analogues give $|Y^{(j)}(u)| = O(T^{1/4+\delta})$ for $u \leq \Theta(T)$, hence $C_k(T) = O(T^{5/4+\delta})$ for every fixed $k \geq 1$.

Therefore, with $k \geq 2$ fixed,

$$\int_{|\mu| > M} |\widehat{Y}_T(\mu)|^2 d\mu \leq \frac{2 C_k(T)^2}{(2k-1) M^{2k-1}}, \quad \rho_T(\{|\mu| > M\}) \leq \frac{C_k(T)^2}{\pi (2k-1) \Theta(T) M^{2k-1}}.$$

Cutoff choice. Set $M = M(T) := T^{1/4}$. Middle off-band: $O_\epsilon(T^{1/4}/T^{1/2}) = O_\epsilon(T^{-1/4}) \rightarrow 0$. Tail, with $k = 10$ (say): $C_{10}(T)^2 = O(T^{5/2+2\delta})$, $M^{2k-1} = T^{19/4}$, $\Theta(T) M^{2k-1} \asymp T \log T \cdot T^{19/4} = T^{23/4} \log T$, giving $\rho_T(\{|\mu| > T^{1/4}\}) = O(T^{5/2+2\delta}/T^{23/4} \log T) = O(T^{-13/4+2\delta}/\log T) \rightarrow 0$.

Hence $\rho_T(\{|\mu| > 1 + \epsilon\}) \rightarrow 0$ for every $\epsilon > 0$.

Weak- limit.* Banach–Alaoglu extracts a weak-* convergent subsequence $\rho_{T_k} \rightharpoonup \rho$. For $\varphi \in C_c(\mathbb{R} \setminus [-1, 1])$ supported in $\{|\mu| > 1 + \epsilon\}$ for some $\epsilon > 0$, $\int \varphi d\rho = \lim \int \varphi d\rho_{T_k} = 0$ (since total off-band mass $\rightarrow 0$ dominates any bounded continuous test function supported there). Hence $\text{supp } \rho \subseteq [-1, 1]$. Total mass in the limit: $\rho([-1, 1]) = \lim \rho_{T_k}([-1, 1]) = \lim(\rho_{T_k}(\mathbb{R}) - \rho_{T_k}(\{|\mu| > 1\})) = 2 - 0 = 2$. Evenness passes from ρ_T to ρ . $\square$

References

- [1] H.M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974 (Riemann–Siegel main formula).
- [2] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann-Siegel-Formel*, Doctoral thesis, Georg-August-Universität Göttingen, 1979 (reissued 2015); establishes $|R_{\text{RS}}(t)| = O(t^{-3/4})$ with explicit constants for the second-order Riemann–Siegel formula.

- [3] E.C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford, 1986 (Hardy–Littlewood second moment, ϑ -asymptotics).